

VEGA **HOME
OF VALUES**

This is VEGA



Headquarters in Schiltach

Innovation Hub
in Karlsruhe

From 2028 onwards,
Kuppenheim location

1,300

employees
in Schiltach



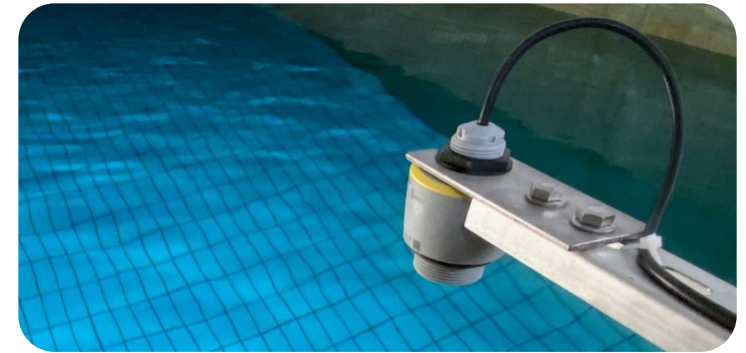
100

IT employees

2,600 employees
worldwide



30 employees
in Karlsruhe





Smart Watering for Urban Trees

Smart Watering for Urban Trees

Hackathon Challenge

How can we use digital technologies to water urban trees more effectively?

Tags:



Sustainability



Smart City



Climate change



IoT



Digitalization

The Problem

Urban trees need water – but key information is often missing

Heat waves and droughts are increasing, making urban trees especially vulnerable.

Cities invest significant effort in watering and often rely on citizens for support.

However, one crucial piece of information is often missing:
Which trees actually need water – and when?



Why does it matter?

Without data, watering becomes guesswork

Possible consequences:

 Trees are watered unnecessarily

 Other trees receive too little water

 Staff and resources are used inefficiently

 Citizen engagement cannot be targeted effectively

The challenge is to enable better data-driven decisions.

Your Mission

Develop an innovative solution

Develop a **simple and scalable** idea that helps to:

- Make the water needs of urban trees or green spaces visible
- Support cities in irrigation decisions
- Engage citizens effectively

Technical framework: ESP32 & capacitive soil moisture sensor

How you create value from this is entirely up to you.



Expected Outcome

Your Pitch

Each team delivers :



A compelling idea with clear benefits



A technical concept



A prototype or mock-up



A 5-minute pitch

Target groups: Cities & municipalities, citizens, environmental and climate initiatives

Evaluation

The jury will primarily evaluate:

- Innovation and creativity
- Practical impact and value
- Technical feasibility
- Scalability
- Pitch quality

Key question: How can soil moisture data help water urban trees more efficiently and make better use of available resources?

